

Security incident report

Section 1: Identify the network protocol involved in the incident

The network protocols identified during the investigation were DNS (Domain Name System) and HTTP (Hypertext Transfer Protocol). DNS was used to resolve the domain names yummyrecipesforme.com and greatrecipesforme.com into IP addresses. HTTP was used to establish web connections and transfer webpage data between the user's browser and the websites. The tcpdump log showed DNS queries followed by HTTP GET requests to both websites.

Section 2: Document the incident

On the date of the investigation, multiple customers reported that yummyrecipesforme.com prompted them to download an executable file claiming to provide access to free recipes. After executing the file, users were redirected to greatrecipesforme.com and experienced reduced system performance.

The cybersecurity team created a sandbox environment and analyzed the network traffic using tcpdump. The packet capture showed a DNS request for yummyrecipesforme.com followed by an HTTP connection to the website. Shortly afterward, the browser initiated a download and then generated a DNS request for greatrecipesforme.com. The DNS server returned the IP address associated with the new domain, and the browser established an HTTP connection with that website.

Further investigation confirmed that the website had been compromised. A former employee gained unauthorized access to the web hosting administrative account through a brute force attack. The attacker successfully

guessed the default administrator password after multiple login attempts. Once access was obtained, the attacker inserted malicious JavaScript into the website source code. The script prompted visitors to download an executable file. Analysis of the downloaded file revealed that it redirected users from yummyrecipesforme.com to greatrecipesforme.com, where malware was hosted. The attacker also changed the administrator password, preventing the website owner from accessing the administrative panel.

Evidence used during the investigation included customer complaints, tcpdump network traffic logs, website source code analysis, and malware file analysis.

Section 3: Recommend one remediation for brute force attacks

One effective security measure is implementing multi-factor authentication (MFA) for all administrative accounts. MFA requires users to provide an additional authentication factor beyond a password, such as a verification code generated on a mobile device. Even if an attacker successfully guesses or obtains a password, they would still need the second authentication factor to gain access. This significantly reduces the likelihood of successful brute force attacks and improves overall account security.